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ChE 210 Quiz #2 Thursday 12 February 2026

To Receive Full Credit, You Must Show All Your Work

The Linde-Frankl separation process uses a feed of atmospheric air (Stream 1) which has a composition of 20.98 mol% oxygen (O_2), 78.08 mol% nitrogen (N_2), and 0.94 mol% argon (Ar) to produce a product (Stream 2) enriched in oxygen ~~and~~ a byproduct (Steam 3) which is enriched in nitrogen and *contains no argon*.

(a) Draw and label a block flow diagram. Define a system. For the 3 streams, use $\dot{n}_i$ for total molar flow rates and x_{i,O_2}, x_{i,N_2}, and $x_{i,Ar}$ for mole fractions for oxygen, nitrogen, and argon, respectively. Use i = 1, 2, & 3.

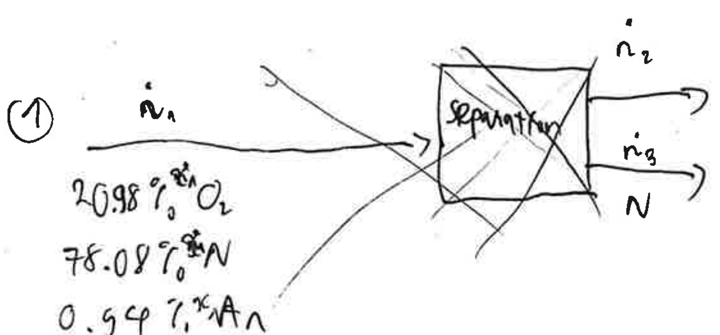
(b) Do a DoF analysis using p. 105 of Murphy. Give values for 4a, 4b, 5a, 5b, 5c, and 5d. Compute the DoF.

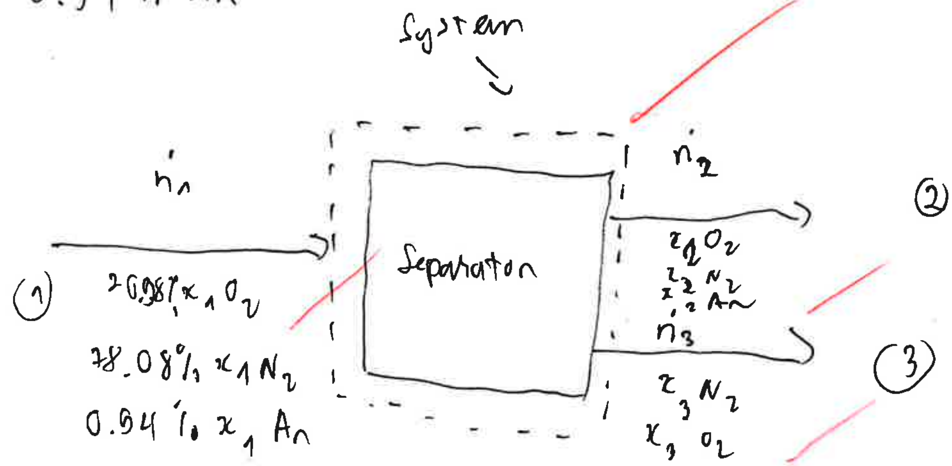
(c) If the DoF < 0, stop here. The problem is overspecified. If DoF = 0, solve the problem. If DoF = 1, set $\dot{n}_2$ = 100 kmol/hr and solve. If DoF = 2, also set x_{2,O_2} = 0.9 and solve. If the DoF = 3, also set x_{3,N_2} = 0.953 and solve. If the DoF = 4, also set $\dot{n}_1$ = 224 kmol/hr and solve. If the DoF = 5, also set $x_{2,Ar}$ = 0.091 and solve. If the DoF = 6, also set x_{3,O_2} = 0.067 and solve.

Steam 3 no Argon

a)

O_2
N_2
Ar





b)

4a = 8

4b = 0

5a: $\cancel{3}$

5b: $\cancel{0}$

5c: $\cancel{0}$ $\cancel{3}$

5d: $\cancel{8}$ $\cancel{0}$

DOF = (4a + 4b) − (5a + 5b + 5c + 5d)

= 8 − $\cancel{6}$ = $\cancel{2}$

$\boxed{\text{DOF} = \cancel{2}}$

c) Because DOF = 2 so we have

$x_2 O_2 = 0.9$ and $\dot{n}_2 = 100$

Argon 2:

$$\dot n_1 (x_1 Ar) = \dot n_2 (x_2 Ar)$$

we have $\dot n_1 = \dot n_2 + n_3$

$$\Leftrightarrow \dot n_1 = 100 + n_3$$

$\Rightarrow$ Oxygen: $\dot n_1 (0.2098) = 100(0.9) + \dot n_3 (x_3 O_2)$

$\Rightarrow$ Nitrogen $=$ $\dot n_1 (0.7808) = 100 (x_2 N_2) + \dot n_3 (x_3 N_2)$

$\Rightarrow$ Argon $=$ $\dot n_1 (9.4 \times 10^{-3}) = 100 (x_2 Ar) + 0$

$\Rightarrow \dot n_1 =$

solve?